

MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

- 1 Real Analysis Lecture 16
 - Continuity
 - Connectedness

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 - Connectedness

Continuity

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$$d_S(x, p) < \delta \Rightarrow d_T(f(x), f(p)) < \epsilon.$$

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Equivalently, $f(B_S(p; \delta)) \subseteq B_T(f(p); \epsilon)$. We say f is continuous on $A \subseteq S$ if $f(x)$ is continuous at every point $p \in A$. If f is continuous on S , we simply say f is continuous.

Sequence version

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Intuitively, continuity means

$$\lim_{n \rightarrow \infty} f(x_n) = f\left(\lim_{n \rightarrow \infty} x_n\right).$$

Compositions

Theorem

Suppose (S, d_S) , (T, d_T) , and (U, d_U) are metric spaces and $f : S \rightarrow T$ and $g : T \rightarrow U$ are functions.

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Suppose (S, d_S) , (T, d_T) , and (U, d_U) are metric spaces and $f : S \rightarrow T$ and $g : T \rightarrow U$ are functions.

If f is continuous at p and g is continuous at $f(p)$, then the composition $h = g \circ f : S \rightarrow U$ defined by $h(x) = g(f(x))$ is continuous at p .

Challenge!

Problem

Prove the previous theorem!

Preimages

Definition

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Suppose $f : S \rightarrow T$ is a function and that $Y \subseteq T$.
The **inverse image** or **preimage** of Y is the set

$$f^{-1}(Y) = \{x : x \in S, \text{ and } f(x) \in Y\}.$$

Challenge!

Problem

Suppose that $f : S \rightarrow T$ and $Y \subseteq Z \subseteq T$.
Prove that

$$f^{-1}(Y) \subseteq f^{-1}(Z).$$

Open sets

Theorem

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Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \rightarrow T$ is a function.

Then f is continuous if and only if $f^{-1}(V)$ is open for all open subsets $V \subseteq T$.

Open sets

Proof.

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$(\Rightarrow)$

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Suppose $V \subseteq T$ is open.

Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.

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Proof.

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Suppose $V \subseteq T$ is open.

Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.

Since V is open, there exists $\epsilon > 0$ such that $B_T(f(x); \epsilon) \subseteq V$.

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Suppose $V \subseteq T$ is open.

Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.

Since V is open, there exists $\epsilon > 0$ such that $B_T(f(x); \epsilon) \subseteq V$.

Since f is continuous at x , there exists $\delta > 0$ such that

$f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon) \subseteq V$.

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Since f is continuous at x , there exists $\delta > 0$ such that

$f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon) \subseteq V$.

Consequently $B_S(x; \delta) \subseteq f^{-1}(V)$.

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Since f is continuous at x , there exists $\delta > 0$ such that

$f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon) \subseteq V$.

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This shows that x is an interior point of $f^{-1}(V)$.

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Proof.

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Suppose $V \subseteq T$ is open.

Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.

Since V is open, there exists $\epsilon > 0$ such that $B_T(f(x); \epsilon) \subseteq V$.

Since f is continuous at x , there exists $\delta > 0$ such that

$f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon) \subseteq V$.

Consequently $B_S(x; \delta) \subseteq f^{-1}(V)$.

This shows that x is an interior point of $f^{-1}(V)$.

Since x was arbitrary, this proves $f^{-1}(V)$ is open. □

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Then for all $\epsilon > 0$, the ball $B_T(f(x); \epsilon)$ is open.

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Proof.

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Suppose that for all open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.

Let $x \in S$.

Then for all $\epsilon > 0$, the ball $B_T(f(x); \epsilon)$ is open.

Therefore the preimage $f^{-1}(B_T(f(x); \epsilon))$ is open.

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Suppose that for all open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.

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Then for all $\epsilon > 0$, the ball $B_T(f(x); \epsilon)$ is open.

Therefore the preimage $f^{-1}(B_T(f(x); \epsilon))$ is open.

It contains the point x , which must be an interior point.

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It contains the point x , which must be an interior point.

This means that there exists $\delta > 0$ such that

$$B_S(x; \delta) \subseteq f^{-1}(B_T(f(x); \epsilon)).$$

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Therefore the preimage $f^{-1}(B_T(f(x); \epsilon))$ is open.

It contains the point x , which must be an interior point.

This means that there exists $\delta > 0$ such that

$$B_S(x; \delta) \subseteq f^{-1}(B_T(f(x); \epsilon)).$$

$$\text{Thus } f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon).$$



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Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \rightarrow T$ is a function.

Then f is continuous if and only if $f^{-1}(A)$ is closed for all closed subsets $A \subseteq T$.

Relatively open sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \rightarrow T$ a function.

If f is continuous on a subset $A \subseteq S$, then for any open subset $V \subseteq T$ there exists an open subset $U \subseteq S$ with

$$f^{-1}(V) \cap A = U \cap A.$$

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$$f|_A^{-1}(V) = \{x : x \in A, \text{ and } f(x) \in V\} = f^{-1}(V) \cap A$$

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is open in the subspace A of S .

This is the same thing as

$$f^{-1}(V) \cap A = U \cap A$$

for some open subset

$$U \subseteq S$$



Compact sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \rightarrow T$ a function.

If f is continuous on a compact subset $A \subseteq S$, then $f(A)$ is compact.

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Then $f^{-1}(V_i) \cap A = U_i \cap A$ for some open subset $U_i \subseteq S$.

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$$f(A) \subseteq \bigcup_{i=1}^n V_i.$$

Extreme value theorem

Theorem

Suppose (S, d_S) is a metric space and $f : S \rightarrow \mathbb{R}$ is a function (to Euclidean space).

If f is continuous on a compact subset $A \subseteq S$, then there exists $a, b \in A$ such that

$$f(a) = \inf f(A), \quad \text{and} \quad f(b) = \sup f(A).$$

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The infimum and supremum of a set are adherent points of that set.

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Therefore, since $f(A)$ is closed $y, z \in f(A)$.

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This means that there exists $a, b \in A$ such that $y = f(a)$ and $z = f(b)$. □

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Characterization of intervals

Theorem

If $I \subseteq \mathbb{R}$ is an interval, then there does not exist two nonempty, disjoint open sets $U, V \subseteq \mathbb{R}$ (Euclidean space), with

$$I \subseteq U \cup V.$$

Connected spaces

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Examples:

- the real line with the Euclidean metric $(\mathbb{R}, d_{eucl})$ is connected
- the real line with the discrete metric $(\mathbb{R}, d_{disc})$ is connected
- euclidean space $\mathbb{R}^n$ is connected
- an interval $I \subseteq \mathbb{R}$ with the Euclidean metric is connected

Intermediate value theorem

Theorem (Intermediate Value Theorem)

Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function with the Euclidean metric.

Then for any value y between $f(a)$ and $f(b)$, there exists $c \in [a, b]$ with $f(c) = y$.